before the beam winked out, Arnold was happily reporting, “All systems go, all systems go, horizon approaching, all systems go, all systems go . . .”

As your excitement subsides, you examine the recorded data. There you find the full details of the shifting laser wavelength. You see that as Arnold fell, the wavelength of the laser signal increased very slowly at first, then faster and faster. But, surprisingly, after the wavelength had quadrupled, its rate of doubling became nearly constant; thereafter the wavelength doubled every 0.00014 second. After 33 doublings (0.0046 second) the wavelength reached 4 kilometers, the limit of your recording system’s capabilities. Presumably the wavelength kept right on doubling thereafter. Since it takes an infinite number of doublings for the wavelength to become infinite, exceedingly faint, exceedingly long-wavelength signals might still be emerging from near the horizon!

Does this mean that Arnold has not yet crossed the horizon and never will? No, not at all. Those last, forever-doubling signals take forever long to climb out of the hole’s gravitational grip. Arnold flew through the horizon, moving at the speed of light, many minutes ago. The weak remaining signals keep coming out only because their travel time is so long. They are relics of the past.[21]

After many hours of studying the data from Arnold’s fall, and after a long sleep to reinvigorate yourself, you embark on the next stage of exploration. This time you, yourself, will probe the horizon’s vicinity; but you will do it much more cautiously than did Arnold.

Bidding farewell to your crew, you climb into a space capsule and drop out of the belly of the starship and into a circular orbit alongside it. You then blast your rocket engines ever so gently to slow your orbital motion a bit. This reduces slightly the centrifugal force that holds your capsule up, and the hole’s gravity then pulls you into a slightly smaller, coasting, circular orbit. As you again gently blast your engines, your circular orbit again gently shrinks. Your goal, by this gentle, safe, inward spiral, is to reach a circular orbit just above the horizon, an orbit with circumference just 1.0001 times larger than that of the horizon itself. There you can explore most of the horizon’s properties, but still escape its fatal grip.

As your orbit slowly shrinks, however, something strange starts to happen. Already at a 100,000-kilometer circumference you feel it. Floating inside the capsule with your feet toward the hole and your head toward the stars, you feel a weak downward tug on your feet and upward tug on your